INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

PUNCHING, SHEARING, NOTCHING AND COPING MACHINE, METAL, MECHANICAL DRIVE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers mechanically driven metalworking machines used for performing punching, shearing, notching, and coping operations on steel bars, structural shapes, and plates of the dimensions and strengths specified herein.

2. CLASSIFICATION. The machine shall be classified and identified by size in accordance with the defining characteristics specified in table I. The size of machine to be supplied shall be specified in the acquisition order (see 7.1(c)).

TABLE I. Product classifications.

Size	Nominal capacity (tons)	Throat depth (inches)
4	50	12
5	50	20
7a	70	15
7b	70	22
8a	90	19
9a	110	19
10a	135	29

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3445

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3. SALIENT CHARACTERISTICS

3.1 Design. Unless otherwise specified (see 7.1(d)), the machine shall accomplish not less than the five operations of punching, notching, coping, flat bar shearing, and structural shape shearing. All stations shall be powered by the same drive. The frame and base of the machine shall be an integral unit designed to rest on a flat surface. The machine shall be manufactured in accordance with American National Standards Institute (ANSI) B11.5, "Ironworkers - Safety Requirements for Construction, Care, and Use" (DoD adopted). Provisions suitable for hoisting the machine shall be provided. The base shall provide a means for securely bolting the machine to the floor. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2 Controls. The press shall be equipped with all the controls required for machine operations. All operating controls shall be located convenient to the operator. All operations of the machine shall be controlled by two independent anti-repeat foot switches, except for punch spotting which may be by hand lever. One anti-repeat foot switch shall control performance of the punch and notcher. The second anti-repeat foot switch shall control bar cutter, shear, and coper operations.

3.3 Drive. The machine shall be provided with a mechanical drive. The drive shall include a drive motor, transmission, clutch, flywheel, drive shaft, connecting rod or cam, slide, and ram.

3.3.1 Drive motor. The manufacturers standard motor provided to the commercial market shall be supplied.

3.3.2 Transmission. The transmission shall consist of a gear or a V-belt system. V-belt systems shall be capable of transmitting the required power without belt slippage. V-belt systems shall have provisions for adjusting belt tension.

3.3.3 Clutch. A clutch or similar controlling device for engaging and disengaging the machine's operating functions shall be provided.

3.3.4 Flywheel. The flywheel shall be a balanced assembly having adequate mass and rotational speed to provide supplementary power to the motor drive while performing work operations.

3.3.5 Drive shaft. The drive shaft shall be of adequate size and material to prevent deflection when subjected to maximum torque. The drive shaft shall be mounted on replaceable anti-friction bearings or bushings of a size, type, and precision class suitable for the application.

3.3.6 Connecting rod and cam. The connecting rod or cam shall be arranged to convert rotary motion to reciprocating motion in driving the rams of the machine. The rod or cam shall be mounted on replaceable anti-friction bearings or bushings of a size, type, and precision class suitable for the application.

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3.3.7 Slide and ram. The slide and ram shall provide reciprocating motion for the machine operations. The ram shall be of solid or box-type construction, with reciprocating action guided by a slide or gib type arrangement. The guide or gib shall be adjustable and replaceable.

3.4 Punch station. The punch station shall consist of a punch holder, die holder, and stripper. The punch station shall be capable of punching holes of the sizes specified (see 7.1(e)). The station shall have a throat depth suitable to accommodate the material size specified (see 7.1(f)). Means shall be provided to lower the punch for positioning over the workpiece.

3.4.1 Punch holder. The machine shall be provided with a punch holder for holding the punch in position under the ram. The punch holders shall be constructed of alloy steel and securely hold the punch in position. The punch holders shall assure alignment of the punches with the dies. The punch holders shall be designed to accommodate standard round and shaped hole punches.

3.4.2 Die holder. The machine shall be provided with a die holder for holding the die in position. The die holder shall be constructed of hardened steel precision ground to securely hold the dies in such a manner to prevent the dies from rotating. The die holder shall be designed to accommodate standard round and shaped hole dies.

3.4.3 Stripper. The machine shall be provided with a stripper for automatically releasing the workpiece from the punch when the punch is withdrawn from the die. The stripper shall permit changing the punch without removing the stripper from the machine.

3.4.4 Punches and dies. When required (see 7.1(g)), punches and dies of the sizes and shapes identified shall be provided. The punches and dies shall be tool steel, hardened, and ground for accuracy. Each punch and die shall be permanently marked for size and identification.

3.5 Notcher station. The notcher station shall be capable of notching triangular cuts in metal to meet the requirements specified (see 7.1(h)). The notcher shall be equipped with an adjustable gauge designed to govern the depth of the cut. A guard shall be provided for protecting the operator when the notcher is operating. The blades of the notched shall be replaceable.

3.6 Coper station. The coper station shall be capable of coping square cuts in metal to meet the requirements specified (see 7.1(i)). The coper shall be provided with a gauge to govern the depth of cut and a protective guard as specified for the notcher. The blades of the coper shall be replaceable.

3.7 Flat bar shear station. The bar shear station shall consist of an upper moveable blade and a lower stationary blade, and shall be capable of shearing the maximum width flat bar stock specified (see 7.1(j)). The camber of the sheared edge of the "drop off" cut shall be 1/32 inch or less per 6 inches of bar width. The shear blades shall be mounted in such a manner that they can be replaced without the use of special tools. The flat bar shear shall be equipped with a stripper or hold down which shall be adjustable to accommodate stock of any dimension within the capacity of the machine. The hold down shall be no more than 4 inches from the shearing plane.

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3.8 Structural shape shear. The structural shape and bar stock shear shall be capable of shearing structural shapes and bar stock of the dimensions specified (see 7.1(k)). Structural angle shapes shall be sheared at 90 degrees and mitered at 15, 30, and 45 degrees, both leg up and leg down. The quality of the "drop off" structural angle cuts shall be such that the two legs of the angle are neither apart by more than 91 degrees, 30 minutes, nor closer together than 88 degrees, 30 minutes. The size and arrangement of the blades shall be in accordance with the manufacturer's standard practice. The blades shall be mounted in such a manner that they can be replaced without the use of special tools. The structural shape shear shall be equipped with an adjustable work piece stripper or hold-down, capable of accommodating stock of any shape or dimension within the capacity of the machine.

3.8.1 Angle iron miter gage. An angle iron miter gage shall be furnished on the structural shape shear to allow shearing at angles of 15, 30, 45, and 90 degrees without the use of layout marks. The miter gage shall be adjustable to accommodate angle iron sizes from 1 by 1 inch up to the size specified (see 7.1(l)) An adjustable stop shall be provided for locating the work in the desired position.

3.9 Power requirements. Electric operated presses shall be wired to conform to National Fire Protection Association (NFPA) 79, "Electrical Standard for Industrial Machinery" (DoD adopted). Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 7.1(m)), each press shall be designed to operate on 3-phase, 60-Hertz (Hz) dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (7.1(n)).

3.10 Motors. Motors shall be the manufacturer's standard motor supplied to the commercial market.

3.11 Standard equipment. All standard equipment normally provided with the manufacturers commercial press shall be furnished.

3.12 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described by the procuring activity (see 7.1(o)).

3.12.1 Work light. When specified, a work light shall be provided. The work light shall be equipped with a protective shield, an on-off switch, and shall be adjustable to reflect the maximum light on the work piece. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.12.2 Hour meter. When specified, the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the non-resetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

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3.13 Size and Capacity. The punching, shearing, notching and coping machine shall meet the size and capacity requirements as specified (see 2 and 7.1(c)).

3.14 Physical size limitations. When specified (see 7.1(p)), the physical size of the press shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the press will fit into its future operating location.

3.15 Marking on instruments, control panels, charts, and plates. All words on the press, including all instruments, control panels, charts, plates, and related attachments, shall be in the English language. Characters shall be marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.16 Data plates. All standard data plates normally provided with the manufacturers standard commercial press shall be furnished. Any additional data plates shall be as specified and fully described by the procuring activity (see 7.1(q)).

3.17 Nameplate. Unless otherwise specified (see 7.1(r)), a nameplate shall be securely attached to each press. The nameplate shall contain the information listed below.

a. Nomenclature
b. Manufacturer's name
c. Machine model designation
d. Serial number
e. Contract number or order number
f. National stock number
g. Date of manufacture

3.18 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, "Preferred Metric Units for General Use by the Federal Government", and all other requirements of this commercial item description are met.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

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6. PACKAGING

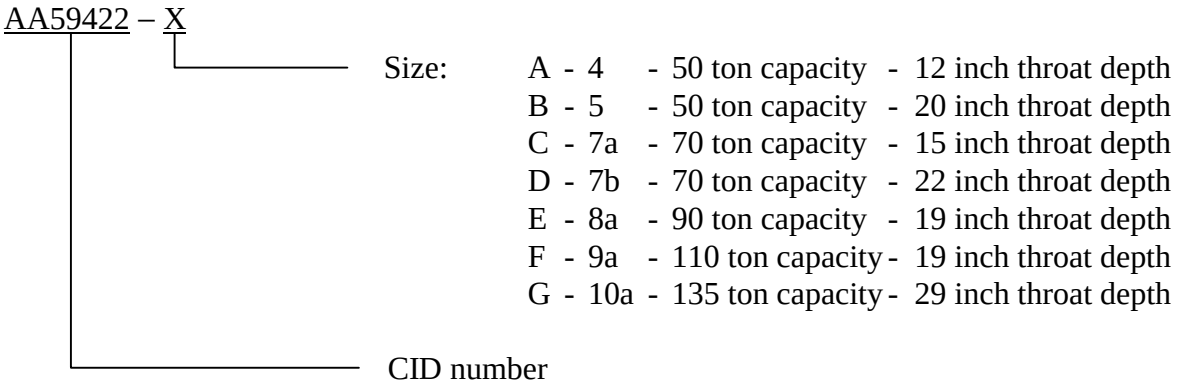
6.1 Preservation, packing, and marking. For acquisition purposes, the machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.1(s)).

7. NOTES

7.1 Ordering data. Acquisition documents must specify the following information:

a. CID document number, revision, and CID PIN
b. Quantity required
c. Size and capacity (see 2 and 3.13)
d. Number and type of work stations, if different (see 3.1)
e. Size of workpiece (see 3.4)
f. Throat depth required, if different (3.4)
g. Punch and die sizes, if required (see 3.4.4)
h. Maximum size of notching cuts (see 3.5)
i. Maximum size of coping cuts (see 3.6)
j. Maximum size of flat bar stock (see 3.7)
k. Maximum size of structural shape bar stock (see 3.8)
l. Maximum angle iron size (see 3.8.1)
m. Electric power requirements, if different (see 3.9)
n. Wiring, as specified (see 3.9)
o. Optional equipment, if required (see 3.12)
p. Physical size limitations, if required (see 3.14)
q. Additional data plate, if required (see 3.16)
r. Nameplate requirements, if different (see 3.17)
s. Packaging requirements (see 6.1)

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



EXAMPLE: AA59422-E would be Size 8a with 90 ton capacity and 19 inch throat depth.

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7.3 National stock number (NSN). The NSNs listed below are assigned for the listed product sizes (see table I). Other NSNs may also correspond with this document.

Assigned NSN	Size
3445-00-266-9574	4
3445-01-174-3755	10A

7.4 Sources for documents.

7.4.1 Government documents. Copies of FAR and federal standards may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328. Electronic copies of federal standards may be obtained from http://astimage.daps.dla.mil/quicksearch/.

7.4.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.4.3 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3445-0153)

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